

* The applied unit values in BenMAP for these measures are county-specific, based on county-specific mean wages, fringe benefits, and/or employment rates. The unit values shown in this table reflect the county average of the county-specific variables applied to these health outcomes.

H.4.1 Non-Fatal Myocardial Infarctions (Heart Attacks)

In the absence of a suitable WTP value for reductions in the risk of non-fatal heart attacks, BenMAP relies on a cost-of-illness (COI) valuation approach. This approach includes two components: (1) direct medical costs and (2) opportunity costs associated with lost earnings due to illness.

The updated valuation function draws on estimates from Maniloff and Fann (2023), who provide age-specific present value estimates of costs associated with non-fatal acute myocardial infarctions (AMIs). These costs include both direct medical expenditures and wage-related opportunity costs and are calculated over the expected remaining years of life at the time of diagnosis, using 2%, 3% and 7% discount rates. The authors provide results for the following age groups: 18–44, 45–54, 55–64, and 65–99. These age-specific values are applied directly in BenMAP to estimate the economic benefits of avoided AMIs.

Maniloff and Fann use household-level data from the Medical Expenditure Panel Survey (MEPS) from 2017–2019, covering over 67,000 individuals, to estimate annual impacts of AMIs on medical expenditures and wage income. Individuals who report having had a myocardial infarction are compared to otherwise similar individuals using a regression framework that controls for age, gender, cancer diagnoses, and other factors. This approach captures not only the direct medical costs of the AMI but also its broader sequelae (e.g., post-event depression or disability) that affect future earnings.

To calculate present value estimates, the authors project average annual cost differences over the estimated number of years lived after diagnosis for each age group. For myocardial infarction, the expected years of life remaining are estimated using either (1) observed years since diagnosis from survey data (conservative estimate), or (2) national life expectancy from the Centers for Disease Control and Prevention (CDC) minus the average age of diagnosis.

To account for local economic conditions, the wage loss estimates from Maniloff and Fann are scaled using county-level wages and labor force participation rates derived from the 2021 American Community Survey (ACS) 5-Year Estimates. Medical and wage-related costs are not assigned to individuals under age 18, consistent with the age coverage in the source data. Table H-13 presents the updated AMI valuation function by age group, disaggregated into total, medical, and wage components.

H.4.2 Minor Restricted Activity Days (MRADs)

Two unit values are currently available in BenMAP for MRADs associated with acute respiratory symptoms. No studies are reported to have estimated WTP to avoid a minor restricted activity day (MRAD). Although Ostro and Rothschild (1989) estimated the relationship between PM_{2.5} and MRADs, rather than minor respiratory-restricted

activity days (MRRADs, a component of MRADs), it is likely that most of the MRADs associated with exposure to PM_{2.5} are in fact MRRADs. The original unit value, then, assumes that MRADs associated with PM exposure may be more specifically defined as MRRADs, and uses the estimate of mean WTP to avoid a MRRAD.

IEc (1993) derived an estimate of WTP to avoid a MRRAD, using WTP estimates from Tolley et al. (1986) for avoiding a three-symptom combination of coughing, throat congestion, and sinusitis. This estimate of WTP to avoid a MRRAD, so defined, is \$38.37 (1990 \$).

Any estimate of mean WTP to avoid a MRRAD (or any other type of restricted activity day other than WLD) will be somewhat arbitrary because the endpoint itself is not precisely defined. Many different combinations of symptoms could presumably result in some minor or less minor restriction in activity. Krupnick and Kopp (1988) argued that mild symptoms will not be sufficient to result in a MRRAD, so that WTP to avoid a MRRAD should exceed WTP to avoid any single mild symptom. A single severe symptom or a combination of symptoms could, however, be sufficient to restrict activity. Therefore, WTP to avoid a MRRAD should, these authors argue, not necessarily exceed WTP to avoid a single severe symptom or a combination of symptoms. The “severity” of a symptom, however, is similarly not precisely defined; moreover, one level of severity of a symptom could induce restriction of activity for one individual while not doing so for another. The same is true for any particular combination of symptoms.

Given that there is inherently a substantial degree of arbitrariness in any point estimate of WTP to avoid a MRRAD (or other kinds of restricted activity days), the reasonable bounds on such an estimate must be considered. By definition, a MRRAD does not result in loss of work. WTP to avoid a MRRAD should therefore be less than WTP to avoid a WLD. At the other extreme, WTP to avoid a MRRAD should exceed WTP to avoid a single mild symptom. The highest IEc midrange estimate of WTP to avoid a single symptom is \$20.03 (1999 \$), for eye irritation. The point estimate of WTP to avoid a WLD in the benefit analysis is \$83 (1990 \$). If all the single symptoms evaluated by the studies are not severe, then the estimate of WTP to avoid a MRRAD should be somewhere between \$16 and \$83. Because the IEc estimate of \$38 falls within this range (and acknowledging the degree of arbitrariness associated with any estimate within this range), the IEc estimate is used as the mean of a triangular distribution centered at \$38, ranging from \$16 to \$61. Adjusting to 2015\$, this is a triangular distribution centered at \$69.58, ranging from \$29 to \$111.

The estimate for the MRADs that is used in EPA benefits analyses can be found in Table H-14.

H.4.3 New Onset Asthma

Maniloff and Fann (2023) developed COI estimates of direct and delayed health care costs, loss of wage income, and home paid caregiving costs associated with new onset asthma. The authors conducted a regression analysis based on 2017 – 2019 medical

expenditure data from the Agency for Healthcare Research and Quality's (AHRQ's) Medical Expenditure Panel Survey (MEPS) to estimate the difference in total medical expenditures between individuals who report having been diagnosed with new onset asthma and otherwise similar individuals who do not report an asthma diagnosis. This approach allows for inclusion of direct medical costs due to the asthma itself and medical costs due to subsequent health effects that may be related to having experienced asthma symptoms. The MEPS data includes actual (not billed) insurer payments, government payments and individual payments, and includes agency and non-agency paid caregiving in its caregiving cost estimates. The authors report the annual costs and present value of lifetime costs of new onset asthma for those ages 18 and older, broken into three cost categories: health care, lost wage income, and home caregiving. The authors estimated lifetime costs using data on expected remaining life expectancy at age of new onset asthma; the resulting present value lifetime estimates are the values used in the BenMAP valuation function. The study estimated the average present value lifetime COI for a new onset asthma to be \$182,681 (2015\$) using a 2% discount rate, \$146,370 (2015\$) using a 3% discount rate, and \$77,490 (2015\$) using a 7% discount rate.

H.4.4 Asthma Symptoms

The valuation estimates for cough, wheeze, chest tightness, and shortness of breath were estimated from the Dickie and Messman (2004) analysis of parents' WTP to relieve asthma symptoms in children and adults. The authors derive the WTP estimates from an attribute-based, stated-choice question assessing preferences to avoid acute illness as part of a survey performed in Hattiesburg, Mississippi in 2000. Survey respondents are asked to identify whether they or their child have experienced the following asthma symptoms in the past year: cough with phlegm, shortness of breath with wheezing, chest pain on deep inspiration, and fever with muscle pain and fatigue. Respondents were then assigned one of sixteen illness profiles varying by symptom, symptom duration, in days, as well as discomfort level. Dickie and Messman (2004) calculate the WTP for children ages zero to seventeen as \$219, for one avoided mild symptom-day (2015\$). The authors also provide WTP estimates by symptom, however, they represent six avoided symptom-days. Therefore, we apply the same WTP value, for one avoided mild symptom-day, to each asthma symptom endpoint.

We calculated the economic value for albuterol use associated with asthma symptoms through prescription costs for albuterol inhalers. Epocrates and GoodRx provide cost and actuation information for four common types of albuterol inhalers in 2020 dollars. Both online resources utilize published price lists, purchases, claim records, and pharmaceutical data to provide clinical statistics. Epocrates and the FDA provide cost and actuation information for one additional, less common, albuterol inhaler.⁷⁰ We divide the cost of inhalers by the actuations per inhaler to calculate an average cost per actuation across all inhaler types. We then adjust the values to 2015\$ using the Consumer Price Index (CPI) for medical care. Since medical cost index data were unavailable for 2020 at the time of these calculations, we used the most recently

available index (2019). The resulting value for asthma symptoms, albuterol use is \$0.35 per actuation (2015\$).

Table H-14 summarizes the unit values utilized by EPA for asthma related health effects.

H.4.5 Allergic Rhinitis

We derived COI estimates for allergic rhinitis (also referred to as hay fever) from the 2005 unit cost data presented by Soni (2008). The study utilizes data from the Medical Expenditure Panel Survey (MEPS) and identifies allergic rhinitis using ICD-9 code 477. Soni (2008) analyzes medical expenditures stratified by age group for the years 2000 and 2005, and calculate the cost-of-illness as the mean expenditures per person for ambulatory care, in-patient services, and prescription medications. The resulting COI for allergic rhinitis is \$600 for ages zero to seventeen (2015\$; Table H-14). The COI estimate represents mean annual medical costs for patients with hay fever. Given that the health impact function for this endpoint relates to allergic rhinitis prevalence, these estimates are more applicable than values representing only first-year costs.

H.4.6 Work Loss Days (WLDs)

Work loss days are valued at a day's wage. BenMAP calculates county-specific mean daily wages from county-specific annual wages by dividing by (52*5), then applies a multiplier of 1.46 for fringe benefits estimates based on the U.S. Bureau of Labor Statistics' 2023 National Compensation Survey and an overhead multiplier of 1.2 from U.S. EPA's National Center for Environmental Economics (NCEE) as described in U.S. EPA (2023) for workers as defined by county-specific employment rate data from the 5-year 2021 American Community Survey. The average of this county-specific daily wage measure across all counties is approximately \$298 for ages 18 to 65 (2015\$, Table H-14).

H.4.7 School Loss Days

We include two costs of school loss days: costs to the caregiver and costs related to the loss of learning for the student not attending school, where the latter are estimated as losses to future lifetime earnings. We calculate each separately and then sum them in the valuation function. We present three alternatives for this function representing three discount rates applied to lifetime earnings portion: 2 percent, 3 percent, and 7 percent.

The caregiver costs assume that an adult caregiver stays home with the child and loses any wage income they would have earned that day. For working caregivers, we follow EPA guidance and value their time at the average wage including fringe benefits and overhead costs (U.S. EPA, 2020a). We apply a multiplier of 1.46 for fringe benefits based on the U.S. Bureau of Labor Statistics' 2023 National Compensation Survey and 1.2 for overhead from U.S. EPA's National Center for Environmental Economics (NCEE) to county-specific mean wage estimates from the U.S. Census Bureau's 2021 5-year American Community Survey (ACS). For nonworking caregivers, we assume that the